```
clear
clc
close all

len = 321;
h = firpm(len-1,[0.005 0.995], [1 1], 'hilbert'); h = h(:);
d = zeros(len,1); d((len+1)/2) = 1;
ht = d + j * h;

freqz(h);
saveas(gcf,'freqz(h)','png');
figure();
freqz(d);
saveas(gcf,'freqz(d)','png');
figure();
freqz(ht);
saveas(gcf,'freqz(ht)','png');

tic
[x, fs] = audioread("fireflyintro.wav");
yr = myconv(x,d);
yi = myconv(x,h);

%put at output of hilbert filter in a for loop
pzold = 0;
fi = zeros(size(yr));
x = zeros(size(yr));
for i = 1:length(yr)
    pz = atan2(yi(i),yr(i));
    fi(i) = (pz-pzold);
    if (fi(i) > pi)
        fi(i) = fi(i) - 2*pi;
    else if(fi(i) < -pi)
            fi(i) = fi(i) + 2*pi;
    end
    end
    pzold = pz;
    x(i) = cos(pz);
end


%gaussian filter
g = gausswin(301,2);
g = g/sum(g);

Cascade = myconv(g,[1 -1]);
freqz(Cascade);
saveas(gcf,'freqz(Cascade)','png');
figure();
```

```
fin = myconv(fi, g);

figure();
NFFT = 2^14;
[S, F, T] = spectrogram(x,hamming(NFFT),round(0.9*NFFT),NFFT,fs);
imagesc(T,F,20*log10(abs(S)),[0 60]);
colorbar;
set(gca,'YDir','normal');
ylim([0 1500])
xlabel('Time [seconds]');
ylabel('Frequency [Hertz]');
title('Spectrogram of the Input x[n]');
saveas(gcf,'firefly_intro','png');
grid on;
hold on
plot((0:length(fin)-1)/fs, fs*fin/(2*pi),"red");
title('Instaneous frequency overlayed');
saveas(gcf,'fi_over_intro','png');
shg

% plot((0:length(fin)-1)/fs, fin);
% ylim([0 1500]);
% xlim([0 10]);
% yticks([200 400 600 800 1000 1200 1400]);
% shg

toc
```